■■ DNS ARMOR: THREAT HUNTING PLATFORM

Incident Detection for DNS Tunneling Attacks

Hybrid Information-Theoretic Entropy, Random Forest ML & Zero-Trust MFA SOC Console

■ **Live Streamlit Cloud Deployed URL:**	**https://dns-tunnel-detector.streamlit.app**
■ **Official GitHub Repository:**	https://github.com/Kamalraj2005/dns-tunnel-detector
■ **Local / Intranet URL:**	http://localhost:8501
■ **Author & Lead Engineer:**	Kamalraj Singh (singhkamalraj5@gmail.com)
■ **Regulatory Compliance:**	NIST SP 800-63B AAL2 • PCI-DSS v4.0 Req 8.4 • MITRE ATT&CK; M1032

1. Executive Summary & Threat Model

The Vulnerability: In modern computer network architectures, **UDP/TCP Port 53 (DNS)** is almost universally allowed outbound across enterprise firewalls to ensure uninterrupted domain resolution. Traditional Intrusion Prevention Systems (IPS) and Next-Gen Firewalls (NGFW) inspect packet headers but fail to evaluate information entropy and lexical characteristics of query subdomains. Adversaries exploit this blind spot to create bidirectional tunnels for covert command-and-control (C2) and exfiltrate sensitive enterprise data.

Common Threat Vectors: Tools like iodine, dnscat2, and Cobalt Strike DNS beacons fragment confidential files (passwords, database dumps, credit cards) into encoded subdomain strings. These queries bypass perimeter segmentation by relaying through recursive corporate resolvers to an attacker-controlled authoritative nameserver.

Mission of DNS Armor: This project delivers an end-to-end incident detection and response system that combines **Shannon Information Entropy**, an **18-feature Random Forest ML model**, automated session correlation, forensic payload reconstruction, and a **Zero-Trust Multi-Factor Authentication (MFA)** console.

2. Zero-Trust Identity & MFA Gateway (RFC 6238 TOTP)

To comply with **NIST SP 800-63B Authenticator Assurance Level 2 (AAL2)** and **PCI-DSS v4.0 Requirement 8.4**, access to operational telemetry and forensic reconstructor data is protected by a two-factor authentication gateway:

- **Factor 1 (Knowledge):** Primary operator credentials with Role-Based Access Control (RBAC).
- **Factor 2 (Possession):** Cryptographic RFC 6238 Time-based One-Time Password (TOTP) supported by Google Authenticator, Microsoft Authenticator, and 1Password.
- **Evaluator Quick-Pass:** A 1-click live dynamic OTP auto-fill button allows rapid viva testing without mobile scanning.

Operator Identity	Assigned Role	Clearance Level	Login Username	Master Password	Base32 TOTP Secret
Alex Mercer	SOC Tier-2 Analyst	`RESTRICTED_INVESTIGATOR`	`analyst@soc.corp`	`CyberDefense@2026`	`JBSWY3DPEHPK3PXP`

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3. System Architecture & Detection Pipeline

The architecture consists of five sequential processing layers operating with sub-millisecond per-query latency:

Pipeline Stage	Component & Subsystem	Technical Specifications & Functions
Stage 1: Ingestion	PCAP Parser & Live Sniffer src/capture/	Reads offline .pcap/.pcapng captures via Scapy (with PcapNgReader fallback). Non-blocking live wire sniffing on UDP port 53 via Npcap driver.
Stage 2: Features	Feature Extraction Engine src/features/	Computes Shannon Entropy H(X), RFC 1035 label lengths, hexadecimal/digit density, vowel ratios, and longest consonant runs across 18 numerical dimensions.
Stage 3: Hybrid Engine	Rules + Random Forest ML src/detection/	Tier 1: 6 fast-path heuristic rules (entropy, oversized labels, covert TXT/NULL records). Tier 2: Supervised Random Forest Classifier (50 estimators, max depth 6).
Stage 4: Forensics	Incident & Payload Decoder src/detection/incident_engine.py	Sessionizes queries by (Client IP, Apex Domain) over sliding windows. Reassembles and decrypts chunked Base64, Hexadecimal, and Base32 leaked payloads.
Stage 5: SOC Console	Streamlit Cyber Dashboard dashboard/app.py	8 interactive glassmorphic operational tabs, radial threat score gauges, Plotly distributions, and CSV/JSON SIEM export facilities.

4. Mathematical Formulation & Shannon Entropy

Shannon Information Entropy Formula: Measures the unpredictability or density of encoded data within a domain string:

$$H(X) = -\sum P(x_i) \times \log_2 P(x_i)$$

- **Natural Language / Benign Domains:** $2.0 \leq H(X) \leq 3.2$ (e.g., mail.google.com, api.github.com).
- **Tunneling / Encrypted Payloads:** $3.85 \leq H(X) \leq 4.8+$ (e.g., b64-dGhpcyBpcyBhIHNlY3Jld...).

Composite Threat Score Formulation:
Query Score = (Heuristic Score × 0.45) + (ML Probability × 100 × 0.55)
Session Incident Threat Score = min(100, (Mean Query Threat × 0.65) + (Behavioral Score × 0.35))

5. The 8 Operational Dashboard Tabs

Tab # & Name	Core Functionality & Analyst Value
Tab 1: ■ Security Incidents	Correlated session cards, radial threat gauges (0-100), MITRE tags, triggered heuristics, and raw query drill-downs.
Tab 2: ■■ Exfiltration Forensics	Automated reassembly of chunked Base64, Hexadecimal, and Base32 data leaks into plain-text recovered files and passwords.
Tab 3: ■ Telemetry & Profiling	Interactive Plotly histograms of Shannon Entropy, Query Length vs Entropy scatter plots, and temporal query velocity.

Tab 4: ■ Live Domain Inspector	Instant analysis sandbox allowing analysts to type any custom FQDN and compute live entropy, 18 lexical metrics, and ML verdict.
Tab 5: ■ AI Engine & Rules	Supervised Random Forest global feature importance rankings explaining exact mathematical weights behind threat decisions.
Tab 6: ■ Query Stream & Export	Searchable live query audit log with 1-click export to CSV and JSON formats for Splunk, Microsoft Sentinel, and SOAR pipelines.
Tab 7: ■ Red Team Attack Lab	In-app attack simulation playground to craft custom confidential payloads and fire real UDP packets on port 53 to test sensors.
Tab 8: ■ Zero-Trust Identity & MFA	Live authenticator device clock health, TOTP synchronization timer, operator access logs, and compliance matrix verification.

6. MITRE ATT&CK;® Alignment & Enterprise Compliance

Technique ID	Technique Name	Tactic	SOC Detection Logic
T1071.004	Application Layer Protocol: DNS	Command & Control	Periodic beaconing and bidirectional TXT/NULL record query abuse.
T1048.003	Exfiltration Over Alternative Protocol: DNS	Exfiltration	High-volume chunked Base64/Hex exfiltration payloads in subdomains.
T1568.002	Dynamic Resolution: DGA	Defense Evasion	High ratio of single-use randomized subdomains generated to evade caching.

7. Local Launchers & Automated Tools (Windows)

Feature / Tool	One-Click Batch Script	Manual PowerShell Command
■ SOC Web Dashboard	`run_dashboard.bat`	`.\.venv\Scripts\streamlit.exe run dashboard/app.py`
■ Real-Time Wire Sniffer	`run_live_monitor.bat`	`.\.venv\Scripts\python.exe live_monitor.py`
■ Live Attack Injector	`run_attack_demo.bat`	`.\.venv\Scripts\python.exe simulate_attack.py`
■ Automated Unit Tests	`run_tests.bat`	`.\.venv\Scripts\python.exe -m unittest tests/test_pipeline.py`
■ Standalone PCAP Analyzer	—	`.\.venv\Scripts\python.exe analyze_pcap.py data/raw_pcaps/mixed_enterprise_traffic.pcap`